

the tens, hundreds, or thousands. By keeping those machines running at peak speed, they could drive down the unit cost of each part and produce cars that were incredibly inexpensive so long as they were completely uniform.

The Japanese car market was far too small for companies such as Toyota to employ those economies of scale; thus, Japanese companies faced intense pressure from mass production. Also, in the war-ravaged Japanese economy, capital was not available for massive investments in large machines.

It was against this backdrop that innovators such as Taiichi Ohno, Shigeo Shingo, and others found a way to succeed by using small batches. Instead of buying large specialized machines that could produce thousands of parts at a time, Toyota used smaller general-purpose machines that could produce a wide variety of parts in small batches. This required figuring out ways to reconfigure each machine rapidly to make the right part at the right time. By focusing on this “changeover time,” Toyota was able to produce entire automobiles by using small batches throughout the process.

This rapid changing of machines was no easy feat. As in any lean transformation, existing systems and tools often need to be reinvented to support working in smaller batches. Shigeo Shingo created the concept of SMED (Single-Minute Exchange of Die) in order to enable a smaller batch size of work in early Toyota factories. He was so relentless in rethinking the way machines were operated that he was able to reduce changeover times that previously took hours to less than ten minutes. He did this, not by asking workers to work faster, but by reimagining and restructuring the work that needed to be done. Every investment in better tools and process had a corresponding benefit in terms of shrinking the batch size of work.

Because of its smaller batch size, Toyota was able to produce a much greater diversity of products. It was no longer necessary that each product be exactly the same to gain the economies of scale that powered mass production. Thus, Toyota could serve its smaller, more fragmented markets and still compete with the mass producers. Over time, that capability allowed Toyota to move